Quality Control Layer 1: Research & Factual Accuracy Audit

Lesson: HTTP Requests and Responses | **Audit Reference:** AUDIT-FACT-001

Overall Audit Status: NEEDS IMPROVEMENT **Factual Accuracy Score:** 87.5% **Issues Detected:** 5

1. Executive Accuracy Assessment

The lesson accurately describes GET/POST semantics, idempotency, and the general meaning of 2xx, 4xx, and 5xx classes per RFC 9110 and MDN. However, it contains several inaccuracies in status code descriptions: 403 is incorrectly tied to authenticated clients, 502 is conflated with 503's overload/maintenance definition, and the hands-on activity sets incorrect expectations for JSONPlaceholder behavior. Minor simplifications exist for 401 and 404. Citations to RFC 9110 and MDN are appropriate, but Source [3] is preview-only and Source [4] is unverifiable from the provided snippet.

2. Factual Claims Analysis & Corrections (5 items)

ID / Severity	Affected Section & Claim	Audit Finding & Supporting Evidence	Recommended Correction
FACT-001 MEDIUM	**Section:** Section 4 - Detailed Explanations (Status Code Categories) *"403 Forbidden: The client is authenticated but lacks permission."*	RFC 9110 Section 15.5.4 defines the 403 Forbidden status code as indicating that the server understood the request but refuses to authorize it. It does not state or require that the client is authenticated. A client may receive 403 without ever authenticating. Describing 403 as exclusively 'authenticated but lacks permission' can mislead learners about when this code applies and conflate it with post-authentication authorization failures. **Evidence:** RFC 9110 Section 15.5.4	**Fix:** 403 Forbidden: The server understood the request but refuses to authorize it. This may occur whether or not the client is authenticated.

FACT-002 MEDIUM	**Section:** Section 4 - Detailed Explanations (Status Code Categories) *"502 Bad Gateway / 503 Service Unavailable: The server is temporarily overloaded or down."*	RFC 9110 Section 15.6.2 defines 502 Bad Gateway as a server acting as a gateway or proxy receiving an invalid response from an inbound/upstream server. Section 15.6.4 defines 503 Service Unavailable as the server being temporarily unable to handle the request due to overloading or maintenance. Grouping 502 with 'temporarily overloaded or down' inaccurately describes 502 and conflates two distinct server-error classes. **Evidence:** RFC 9110 Sections 15.6.2 and 15.6.4	**Fix:** 502 Bad Gateway: The server, while acting as a gateway or proxy, received an invalid response from an upstream server. 503 Service Unavailable: The server is temporarily overloaded or undergoing maintenance.
FACT-003 MEDIUM	**Section:** Section 7 - Hands-on Activities (Activity: Inspect Live HTTP Traffic, Step 6) *"Deliberately send invalid JSON in the console and observe a 400 or 500 response."*	The activity directs learners to use https://jsonplaceholder.typicode.com/posts, a public test placeholder API. JSONPlaceholder generally accepts POST payloads without strict validation and returns 201 Created regardless of body validity; it does not reliably return 400 or 500 for invalid JSON. Telling learners to expect a 4xx or 5xx from this specific endpoint sets incorrect expectations and may confuse them when the observed behavior does not match. **Evidence:** Behavior of JSONPlaceholder test API (public documentation/community knowledge); not verifiable via provided sources [1]-[4]	**Fix:** Use a local development server or a real API that validates request payloads, or adjust the expectation to observe how the specific test endpoint handles malformed input (which may still return 201).
FACT-004 LOW	**Section:** Section 4 - Detailed Explanations (Status Code Categories) *"404 Not Found: The requested resource does not exist."*	RFC 9110 Section 15.5.5 states that 404 indicates the origin server did not find a current representation for the target resource or is not willing to disclose that one exists. Stating simply that the resource 'does not exist' omits the possibility that the server is deliberately withholding confirmation of existence, which is a nuance relevant to security and API design. **Evidence:** RFC 9110 Section 15.5.5	**Fix:** 404 Not Found: The server did not find a current representation for the target resource, or is unwilling to disclose that one exists.

FACT-005 LOW	**Section:** Section 4 - Detailed Explanations (Status Code Categories) *"401 Unauthorized: Authentication is required or failed."*	RFC 9110 Section 15.5.2 defines 401 Unauthorized as the request not being applied because it lacks valid authentication credentials for the target resource. The word 'failed' may imply the server attempted and rejected authentication, which is not necessarily the case; 401 is about missing or invalid credentials, not a failed authentication attempt after credentials were provided. **Evidence:** RFC 9110 Section 15.5.2	**Fix:** 401 Unauthorized: The request lacks valid authentication credentials for the target resource.

3. Verification Scope & Limitations

Source [3] (HTTP: The Definitive Guide) is accessible only as metadata/preview, so full-text claims attributed to it could not be verified page-by-page. Source [4] (YouTube video) is represented only by a snippet; video content and transcript were not provided for full-text verification. Technical verification was therefore performed primarily against RFC 9110 and MDN Web Docs.

4. Conclusion & Certification

Overall the lesson is factually sound in its core concepts but requires correction of specific status code definitions (403, 502, 503) and the JSONPlaceholder activity expectation before it can be considered fully accurate.